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Traditional Bengali Food Classification and Calorie Measurement Using Machine Learning

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Abstract

Food classification and calorie measurement are turning into fundamental concepts that researchers are applying various techniques in different cuisines to analyze dietary patterns, investigate the nutritional composition of foods, and to calculate total caloric intake. This research paper presents a comprehensive approach for classifying and measuring the calorie content of traditional Bengali dishes using machine learning techniques. A custom dataset containing 2000 images of food items is compiled and annotated for both training and evaluation purposes. YOLOv8 and YOLOv7 architecture are used for both object detection and image segmentation which enable accurate identification and localization of food items within images. Additionally, a mathematical model is created to estimate the volume and calories of the detected food items. The proposed method utilizes the power of deep learning algorithms to effectively classify different Bengali dishes, facilitating their recognition and differentiation. Using YOLOv8, the system is able to achieve robust object detection and segmentation, accurately separating various food items within images. As people are becoming more health-conscious, accurate assessment of dietary caloric intake is becoming significant and promising for leading a healthy life. Our research provides a solution for measuring the caloric intake of the food items individuals consume. Our proposed datasets and models which are trained and evaluated for calculating volume and calorie measurements have the ability of achieving a high level of accuracy and performance. This research provides a valuable foundation for the development of a practical and efficient system for calorie measurement of traditional Bengali dishes, potentially contributing to healthy eating habits and facilitating diet planning.

Keywords: Food Analysis; Bengali Cuisine; Classification; Object Detection; Segmentation; Dietary Assessment.

1. Introduction

Food category classification is a vital technique used in several food-related functions, including monitoring healthy eating habits, computational cooking, and meal recommendation systems (Wang et al., 2015a). The field of automatic food recognition has gained more attention due to our growing emphasis on healthier lifestyles (Chen et al., 2012). In recent years, there has been an increasing focus on machine learning models for identifying and estimating the quantity of food. People typically describe the identification problem as the challenge of categorizing or classifying images and estimating calories. Chronic diseases, including diabetes, obesity, and cardiovascular disease, are leading causes of death and morbidity worldwide, and have lately become epidemics in many Asian nations (Organization of Canada, 2005). Therefore, it is imperative to precisely assess the dietary intake in order to manage and regulate it (Zhu et al., 2010). Technology-inspired solutions to avoid sickness caused by poor diet management are crucial for enhancing food nutrition knowledge. Patients and dietitians have developed various nutrition-measuring methods and applications to help them manage obesity and health hazards (Ocay et al., 2017). In accordance with technological

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trends, food recognition models that extract nutritional information from photographs of captured food have made a significant positive impact on individuals' ability to regulate their diets and health. Food intake appraisal and monitoring are critical components in the management of obesity and various other health-related concerns (Kong, 2012). To get nutritional and caloric information, images of meals can be readily shot and transferred to a central server, replacing the need for a questionnaire or diary (Kitamura et al., 2009). According to the most recent study on Bangladesh, the people like a wide range of tasty foods, snacks, and sweets (Jeny et al., 2019a) (Jeny et al., 2019b) (Hossain et al., n.d.). Bangladeshis are well-known for their hospitality, most of which stems from the distinctive tests of traditional delicacies (Uddin et al., 2021). Bengali cuisine, a distinct blend of rich and varied flavors, is shaped by location, climate, and history. Bengali cuisine emphasizes fresh ingredients and balanced nutrition, making it both delicious and nutritious. However, eating a well-balanced diet and keeping track of your calorie intake are essential for living a healthy lifestyle. Machine learning technologies are becoming more popular in a variety of industries, and deep learning approaches can help improve dietary habits and measure calorie consumption. This study presents a novel approach to identify traditional cuisine in Bangladesh by employing deep learning techniques, including estimated volume estimation and object detection models. The dataset comprises approximately 2000 image representations of five traditional cuisines from Bangladesh. In the domain of image classification, YOLOv8, a mathematical approach, has demonstrated superior performance to alternative methods. Aiming to increase the nutritional value and precision of Bengali dishes while promoting a healthier lifestyle is the objective of the proposed method.

2. Related works:

Food Image Classification is a relatively new subject among the emerging applications of deep learning developments (Uddin et al., 2021). However, few studies have identified traditional foods (Tasnim et al., 2020). In 2017, researchers suggested using Convolutional Neural Networks (CNN) for food classification (Attokaren et al., 2017). The Food-101 dataset was utilized, comprising 101,000 photos and 101 categories. Initially, SVM, Neural Networks and the ResNet18 were employed resulting in accuracies of 50.76%, 56.40% and 67.23% respectively. Then finally CNN was applied to improve the accuracy of nonlinear data which can accurately map non-linear relationships with 86.97% precision.

A group of researchers presents UPMC Food-101, an extensive multimodal dataset that was collected in an uncontrolled setting with significant variation among instances. The UPMC Food-101 dataset has almost 100,000 photos and corresponding descriptions for 101 different food categories. This dataset not only has a large number of images, but also includes food categories of ETHZ Food-101, a well-known public food image resource. This article provides an explanation of the similarities and differences among these datasets. (Wang et al., 2015b).

Another research article appeared in 2016 titled Food vs. Non-Food Classifier. UNICT-FD889 was one of the datasets used in this paper that includes around 4805 food and 8005 non-food photos. The Alex Net model with binary SVM classifier was utilized in this paper with an accuracy of 94.86% (Jeny et al., 2019c)

A method for dietary assessment was proposed by the researchers in 2015. The technique employs a range of images obtained during controlled and natural eating scenarios to automatically detect and localize the presence of food. This is achieved by integrating two concepts: by using global and local features, a group of segmented objects may be partitioned into object classes that are perceptually similar; and these perceptually similar object classes can be used to assess the efficiency of image segmentation (Zhu et al., 2014).

The "Snap, eat, repEat" study presents an additional dataset comprising around 150,000 images of prepared meals and 500 distinct food categories. The objective of this dataset is to develop a food recognition system that can be utilized in dietary assessment and logging applications. In addition, deep learning was employed to quantify food calories, thereby facilitating the development of a user-friendly, intelligent, and precise food recognition system that provides more practical methods for monitoring caloric consumption (Pouladzadeh et al., 2016). NutriNet is an image detection and recognition system for food and beverages based on a convolutional neural network (Mezgec & Korouˇsi´c Seljak, 2017). Four unique deep convolutional neural network architectures were utilised, including Nutrinet, a uniquely designed architecture for this research. The recognition task involved evaluating four different deep learning architectures (NutriNet, AlexNet, ResNet, GoogLeNet) and three types of solvers (AdaGrad, NAG, and SGD). The challenge requires participants to accurately classify images into 1000 distinct categories. Three architectures served as viable options for training on images of food and beverages, as well as a comparison with the NutriNet architecture. Further differentiation from the majority of previous studies is the practical application of the food and beverage image recognition system, which is currently being employed to evaluate the dietary habits of patients diagnosed with Parkinson's disease.

Tasnim et al. (Tasnim et al., 2020) proposed a transfer learning technique to retrain the Inception-v3 model using TensorFlow on a dataset consisting of 7 food items: Kalavuna, Khichuri, Murgir-Roast, Panta-Ilish, Sorse-Ilish, Gorur-Rezala, and Tehari. They developed a successful food item recognition model with a brief training period and attained

improved accuracy.

Uddin et al. (2021) designed a dataset called the Traditional Bengali Food Image (TBFI) for classifying Bengali food images. The dataset consists of 2,835 images of traditional Bengali cuisine, categorized into seven groups: Khichuri, Kala Vuna, Biryani, Dim Vhuna, Fuchka, Panta Ilish and Roshogolla. In an effort to categorize the seven separate classes of their TBFI dataset, they intended to create a whole new CNN model. Subsequently, the researchers utilized a transfer learning technique on the TBFI dataset by employing the VGG16, a pretrained model, to establish a comparison with their novel model. On average, this method was able to identify seven distinct traditional Bengali meals with a precision of 95.2%.

Jeny et al. (Jeny et al., 2019c) proposed an additional CNN model that is based on a deep residual neural network. The fundamental architecture of the Deep Residual Neural Network (DRNN) continues to consist of numerous layers connected via a shortcut known as the identity block. However, an alternative CNN model lacks a shortcut connection. Therefore, the Deep Residual Neural Network is distinct from others in this regard. The authors presented a CNN model that incorporates MobileNet and Inception-V3 in order to detect the local food of Bangladesh. Upon utilizing this model, they discover that the test's accuracy is exceptionally high at 98.16 %. Additionally, they observe that the training accuracy for distinguishing local food is significantly greater than other CNN models. However, MobileNet and Inception-V3 have also performed on this dataset.

Ming et al. (2018) developed a mobile-based dietary tracking system called DietLens, which utilizes a deep-based food image recognition model. They conducted experiments in a laboratory setting using Singaporean food, which is known for its diverse culinary influences from Malay, Chinese, Indian, and Western cultures (Lau et al., 2016). Subsequently, they implemented the recognition model in their prototype system, known as DietLens, and proceeded to evaluate its performance by testing it on actual food photographs captured by users. Additionally, they performed a user study that evaluates the usability of various widely-used food monitoring strategies and their prototype system, specifically focusing on the techniques used for entering food information.

By expanding the conventional UECFood-100, Ege et al. (Ege, Shimoda, & Yanai, 2019) have chosen to generate a food image dataset that has been annotated using instance-based bounding boxes and segmentation masks. 10,000 images from the UECFood-100 were manually annotated with instance-based bounding boxes. By this method, Food calories can be estimated by analyzing the food area in a photograph. The food area is determined by combining estimated and actual scales of food regions.They create a new food image dataset for food detection and segmentation by enhancing the existing UECFood-100 dataset with bounding boxes and segmentation masks.

In conclusion, the field of Bengali food classification has seen limited research, but this paper represents an opportunity to introduce a fresh perspective and contribute to the advancement of this particular domain. The development of this dataset represents a significant contribution to the study of Bangladeshi Traditional Cooked Food with its collection of 2000 images across five different classes, obtained from online platforms and real-life sources. This dataset offers a comprehensive and diverse resource for research, analysis, and application development in the field of Bengali cuisine. By leveraging machine learning technologies like CNN, Inception v3, Tensor Flow, YOLOV8 model and other advanced methods, this research will continue to advance the field of food image classification and calorie measurement. Its main components are data pre-processing, food recognition and segmentation, calorie estimation, and evaluation of the results.

3. Dataset

The Bangladeshi Traditional Cooked Food dataset was created to fill the gap in accessible data regarding this particular cuisine. The collection contains 2000 photos categorized into five classes: rice, fish, meat, vegetables, and biriyani. Chef Md Sumon Bhuiya from Bhuiya Hotel and Restaurant in Dhaka, Bangladesh, reports that these five meals are the top sellers. The dataset is collected from internet platforms like Google, Facebook, and Instagram, as well as real-life sources to increase authenticity. Integrating photographs from online platforms and real-life sources enhances the dataset by providing a variety of views for scholars and practitioners to assess traditional Bengali dishes. Having a wide range of sources helps provide a more thorough portrayal of the cuisine and its cultural importance. The dataset is a great resource for a range of applications, including computer vision, machine learning, and culinary research. Researchers can use the dataset for applications including food recognition, dish classification, ingredient analysis, and recipe development. The collection can also be used to create intelligent food recommendation systems, promote cultural preservation efforts, and build instructional resources focused on Bengali cuisine. Ultimately, creating this dataset is a substantial addition to the research on Bangladeshi Traditional Cooked Food. The dataset contains 2000 photos categorized into five types, providing a wide-ranging and varied resource for study, analysis, and application development related to Bengali cuisine.

4. Methodology

The study of machine learning techniques for analyzing food images and estimating their caloric content is gaining popularity, with applications in dietary tracking, health management, and personalized nutrition. Accurately determining

calorie content from food images can be challenging due to their complex nature. This methodology presents a systematic approach for estimating calorie content from food images, involving data pre-processing, food recognition and segmentation, calorie estimation, and evaluation of results. To train a deep learning model, YOLOv8 and YOLOv7 models are used. In this paper, Bengali cuisine images of five different types are used for the implementation. Large images require more memory and processing power, requiring various preparations and techniques for augmentation. Feature extraction is performed using YOLOv7 and YOLOv8, and a classifier is applied to classify images using a "SoftMax" activation function-based classifier. The YOLOv8 model has better performance than YOLOv7 for the classification of the food data, segmenting the data into five classes, and generating a ".txt" file with the segmented data.

There have been several models proposed for the detection of food images, measuring the number of calories present in food items as well as analyzing the calorie intake of a person by determining their daily dietary information. Several methods and algorithms have been implemented in various aspects of these concerns. Preprocessing is crucial for efficient testing of images which are collected from various sources. It means the image size may not be small enough for the model to perform quickly. To resolve this issue, the dataset has been resized to 640×640 pixels for consistency. To increase accuracy, unwanted and duplicate data have been removed. Annotation is required to utilize YOLOv8 and YOLOv7 for training and testing, as the dataset is not annotated, Roboflow is the optimal tool for this purpose.The dataset is split into a 60:20:20 ratio for training, validation, and testing, with 5 different classes and labeled using Roboflow to minimize data redundancy. By utilizing Roboflow, the model can effectively train and test models, ensuring accurate and consistent results.

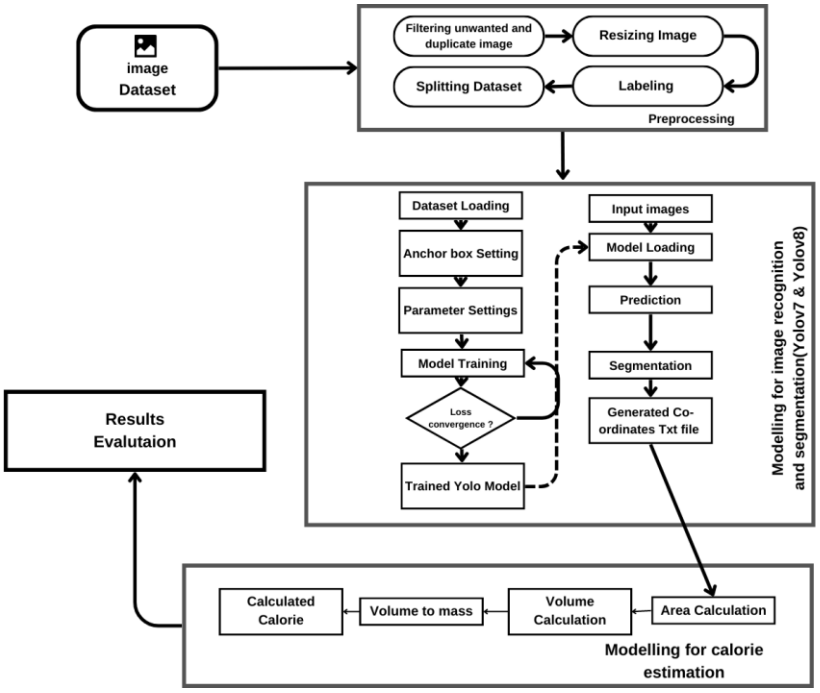


Figure 1: Steps of methodology

Researchers construct a large food data set specifically for Bengali food. The data set is crawled from Google, Instagram, and Facebook by using the food name as a query. The data set contains 5 food categories and their caloric information.

4.1 Food recognition model:

4.1.1. Yolov7

YOLOv7 (C.-Y. Wang, Bochkovskiy, & Liao, 2023) is a single-stage real-time object detector. It was launched to the YOLO family in July 2022. According to the YOLOv7 study, it is the fastest and most accurate real-time object detection yet. YOLOv7 set a significant benchmark by measuring its performance on a different scale.

The E-ELAN is the computing component of the YOLOv7 backbone. It draws inspiration from past research on

network efficiency. It was created by considering the following aspects that influence speed and accuracy.

4.1.2. Yolov8

YOLOv8 (ultralytics, 2024) is a cutting-edge object detection model that uses a deep neural network to identify and classify objects in real time. It is an updated version of YOLOv7, with better accuracy and speed. Memory access costs. YOLOv8 incorporates a new backbone network architecture known as CSPDarknet, which is one of its most significant enhancements. Another enhancement to YOLOv8 is the usage of a new loss function called Focal Loss. Overall, YOLOv8 is a highly efficient object detection model that provides cutting-edge accuracy and speed, making it ideal for real-time applications.

4.2 Calorie measurement model:

The quantification of calories based on volume is essential for comprehending the nutritional composition of different food items. By measuring the volume of a food item and using appropriate conversion factors, it is possible to determine the number of calories associated with that volume. The first step for the calorie measurement model is segmentation. Segmentation in computer vision involves breaking down an image into distinct regions or segments, which is essential for various applications. Both YOLOv7 and YOLOv8, typically known for object detection, can also be adapted for segmentation. The process starts with preparing a dataset where images are annotated at the pixel level to define the boundaries of different segments or objects. Next, the model is trained on these annotated images, where it learns to predict segmentation maps by minimizing the differences between its predictions and the ground truth through a loss function. This training demands substantial labeled data and computing power. Once trained, the model can segment new images in real-time, applying post-processing like thresholding to enhance the segmentation quality.

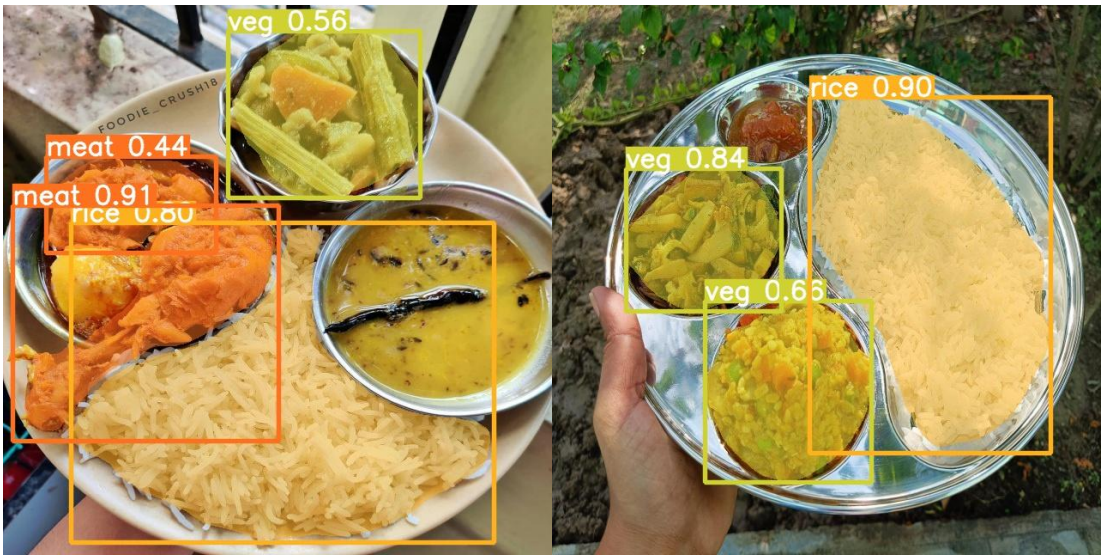


Figure 2: Sample of segmented images

In geometry, the polygon rule is a method used to measure the area of polygons by breaking them down into smaller, manageable parts like triangles or rectangles, and then calculating the area of these parts. By adding up the areas of all parts, the total area of the original polygon is obtained. This rule is beneficial in fields like architecture and geography where precise area measurements are critical. For three-dimensional objects, once the area is calculated using the polygon rule, the volume can be estimated by multiplying the area with the object's height, assuming a uniform height throughout. This is particularly useful in various scientific and engineering applications. In the context of nutrition, understanding the volume of food items is crucial for estimating their caloric content. By measuring the volume and applying relevant conversion factors, the caloric value of the food can be estimated, which is vital for dietary management and nutritional analysis.

5 Result Analysis

This research presents a comparative analysis of the performance of YOLOv7 and YOLOv8 object detection models

in the context of Bengali food recognition and calorie estimation of the polygon method. The models were evaluated based on several performance metrics, including accuracy, precision, recall, and F1 score. Additionally, the effectiveness of a proposed method for estimating food calorie content based on volume measurements is also evaluated.

A. Performance Metrics Equations:

The performance metrics used to evaluate the models are given below:

Accuracy: Accuracy refers to the ratio of accurate predictions in the confusion matrix to the total number of predictions made.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Precision: Precision refers to the ratio of accurate predictions in the confusion matrix to the total number of positive predictions.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

Recall: Recall, often referred true positive rate, quantifies the ratio of accurately anticipated positive instances to the total number of actual positive instances.

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

F1 Score: The balanced average of precision and recall is a single statistic that effectively balances both requirements.

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

B. F1 Confidence Curve

The F1 confidence curve (Lt, 2021) is a graphical representation that illustrates the correlation between the F1 score and the confidence threshold employed in a binary classification problem. This curve is particularly beneficial in imbalanced datasets, where the F1 score is a robust performance metric, as it demonstrates the process of how precision and recall alter as confidence thresholds are adjusted.

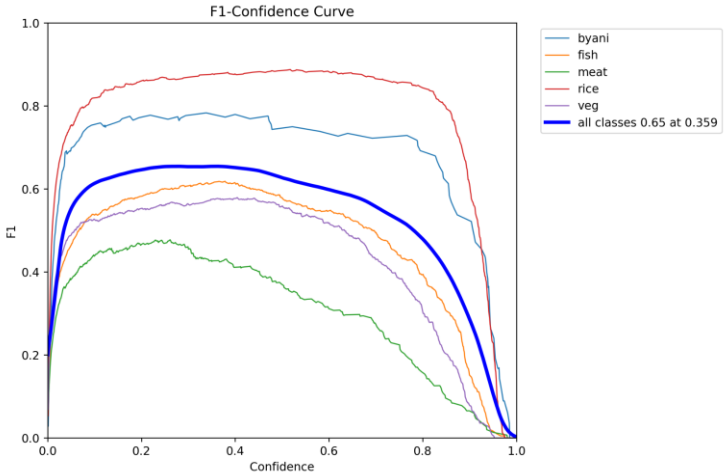


Figure 3: F1-Confidence Curve

C. Precision-Recall (PR) Curve

The PR curve (Lt, 2021) demonstrates the trade-off between precision and recall at various thresholds. Precision evaluates the model's capacity to identify true positives, while recall evaluates its ability to prevent false positives. The curve is especially beneficial for assessing models on imbalanced datasets, as it highlights the model's ability to predict positive instances.

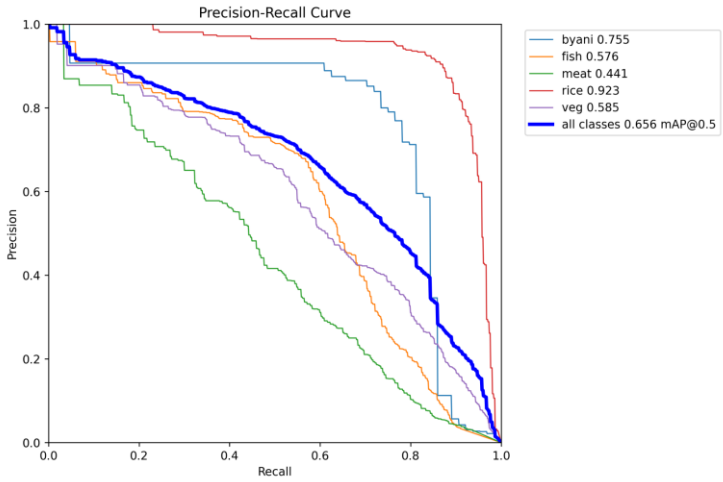


Figure 4: Precision-Recall Curve

D. Confusion Matrix

The confusion matrix offers a comprehensive analysis of the model's prediction outcomes by comparing the anticipated classes with the actual classes. It comprises:

A true positive (TP): Refers to the accurate prediction of positive situations.
A True Negative (TN): Refers to events which are correctly anticipated as negative.
A false positive (FP): Refers to the incorrect prediction of positive instances.
A false negative (FN): Refers to situations that are incorrectly anticipated as negative
.

By utilizing the confusion matrix, we can extract multiple performance measures that offer vital information about the model's capability.

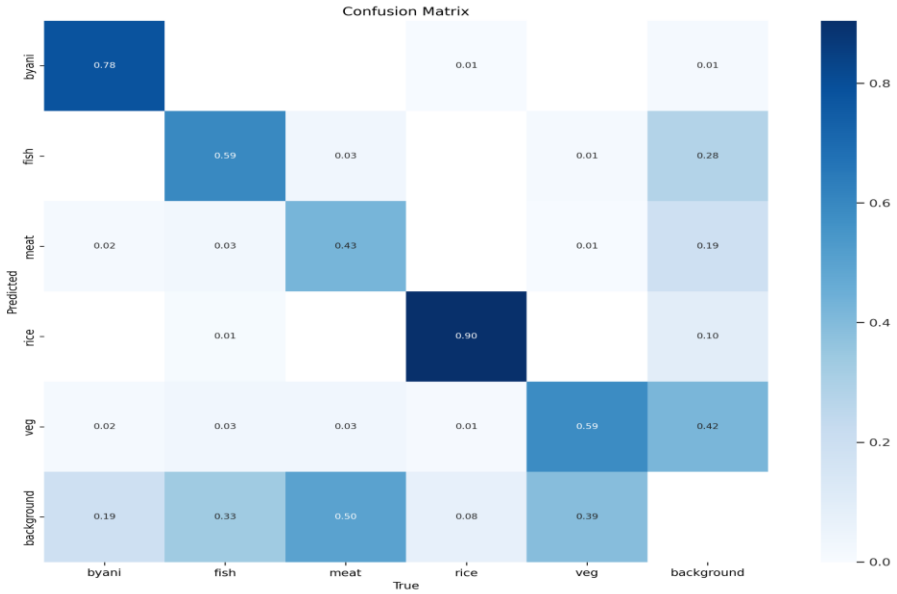


Figure 5: Confusion Matrix

5.1 Results of YOLOv7 and YOLOv8

The evaluation of YOLOv7 and YOLOv8 on Bengali food recognition and calorie estimation is summarized in the following tables:

Table 1: YOLOv7 Results

Food Type	Accuracy	Precision	Recall	F1 Score
Biriyani	0.91	0.93	0.56	0.70
Fish	0.87	0.62	0.55	0.58
Meat	0.81	0.40	0.28	0.33
Rice	0.91	0.74	0.75	0.72
Veg	0.75	0.34	0.53	0.42
Average	0.85			

Table 2: YOLOv8 Results

Food Type	Accuracy	Precision	Recall	F1 Score
Biriyani	0.96	0.98	0.77	0.86
Fish	0.88	0.65	0.60	0.62
Meat	0.86	0.63	0.43	0.51
Rice	0.96	0.89	0.90	0.90

Veg	0.85	0.53	0.59	0.56
Average	0.90			

5.2 Analysis of YOLOv8 vs YOLOv7

The research findings indicate that YOLOv7 demonstrates greater efficacy compared to YOLOv8 in terms of accuracy, precision, recall, and F1 score for all assessed food categories. It is worth mentioning that YOLOv8 attained an average accuracy of 90%, which is considerably higher than the average accuracy of YOLOv7, which was 85%. The improved performance of YOLOv8 can be attributed to multiple factors:

1. Model Architecture: The YOLOv8 model architecture has modifications that improve its ability to extract features, enabling more accurate identification of intricate patterns in Bengali food images.
2. Training Data: While both models are trained using the same dataset, YOLOv8 utilized a more extensive and diverse set of Bengali cuisine attributes, resulting in higher data quality and capturing a broader range of features.
3. Model Optimizations: YOLOv8 achieved higher efficiency metrics compared to YOLOv7 due to the implementation of improved hyperparameter tuning and optimization approaches.

5.3 Results for Calorie Measurements

A regression model was employed to evaluate the proposed method for estimating the calorie content of food based on volume measurements. The model was trained on a dataset comprising food items with known calorie and volume values. The study employed a polygon model for volume estimation to compare calorie estimations that are derived from both mass and volume measurements. Key findings consist of:

- The average accuracy of volume estimation was 77.94%, with a range of 64.65% to 91.78% across a variety of food categories.
- The algorithm's efficacy for practical applications in dietary monitoring and calorie counting was demonstrated by its substantial potential for accurate calorie estimation based on volume measurements.

Test	Food Type	Estimated Volume	Predicted Volume	Accuracy (\%)
Test 1	Rice	675	569	84.29
	Fish	120	93.685	78.07
Test 2	Rice	225	291.54	77.17
	Meat	60	82.5	72.727
	Vegetable	75	62	82.66
Test 3	Biriyani	365	238.8	65.205
Test 4	Biriyani	730	472.0	64.65
Test 5	Rice	450	515.06	87.36
Test 6	Biriyani	365	415.63	87.818
	Vegetable	75	66.42	88.56
Test 7	Biriyani	365	432.0	84.49
Test 8	Fish	120	143	83.92

	Meat	60	98.8	60.728
	Rice	562.5	516.23	91.78
Test 9	Rice	225	110.86	49.27
	Fish	60	82	73.17
Test 10	Biriyani	365	391.75	93.17

In summary, the research suggests that YOLOv8 is significantly more effective than YOLOv7 in Bengali Food Recognition, and the polygon method has advantages for calorie estimation tasks. The improved efficiency of YOLOv8 is a result of its optimized training processes, richer training data, and architectural advancements. Furthermore, the proposed method for calorie estimation, exhibits promising results, indicating its potential to accurately evaluate the volume from which the calorie is measured.

6 Conclusion

This research study presents a machine learning approach to categorize and analyze the calorie content of traditional Bengali cuisine. Using a customized dataset consisting of 2,000 food photos, we utilize YOLOv8, a state-of-the-art algorithm, for precise detection and segmentation of objects. Additionally, we implement a mathematical model to ensure perfect measurement of volume and calculation of calories.The results of our research validate the efficacy of this method in accurately classifying Bengali cuisine and accurately assessing the caloric content of those items. Despite the primary focus on Bengali cuisine, we could potentially apply the designed techniques to other cuisines.Overall, this approach efficiently facilitates the selection of healthier food options and improves comprehension of nutrition. Subsequent research should emphasize the dataset's expansion, improved food volume estimation, caloric calculation, and dietary item diversification. This will enhance the accuracy and usefulness of the suggested approach in various dietary contexts.

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